

MATH 2211, SPRING 2026: HOMEWORK 5

- (1) Suppose that $f : X \rightarrow Y$ and $g : Y \rightarrow X$ are functions such that $g \circ f = \text{id}_X$. Show that f is injective (that is, one-to-one) and that g is surjective (that is, onto). Show by example that neither f nor g need be bijective.
- (2) Suppose that $T : V \rightarrow W$ is an invertible F -linear transformation of F -vector spaces. Show that the inverse function $T^{-1} : W \rightarrow V$ is also a linear transformation.

We will prove the following result in class:

Proposition 1 (Rank-Nullity). *If $T : V \rightarrow W$ is a linear transformation of F -vector spaces, then*

$$\dim \ker(T) + \dim \text{im}(T) = \dim W.$$

- (3) Decide if the following can exist. Justify your response by explaining why it cannot or by giving an example when it does:
 - (a) A linear transformation $T : F^4 \rightarrow F^4$ with $\ker(T) = \text{im}(T)$.
 - (b) A linear transformation $U : F^5 \rightarrow F^5$ with $\ker(T) = \text{im}(T)$.
- (4) Suppose that $S, T : F^n \rightarrow F$ are linear transformations with the same kernel. Show that there exists $c \in F^\times = F \setminus \{0\}$ such that $cS = T$.
- (5) Decide if the following are true for a linear transformation $T : V \rightarrow W$:
 - (a) If T is injective and $v_1, \dots, v_n \in V$ are linearly independent, then $T(v_1), \dots, T(v_n) \in W$ are also linearly independent.
 - (b) If T is surjective and $v_1, \dots, v_n \in V$ are linearly independent, then $T(v_1), \dots, T(v_n) \in W$ are also linearly independent.
 - (c) If T is injective and $v_1, \dots, v_n \in V$ span V , then $T(v_1), \dots, T(v_n) \in W$ span W .
 - (d) If T is surjective and $v_1, \dots, v_n \in V$ span V , then $T(v_1), \dots, T(v_n) \in W$ span W .
- (6) Suppose that $T : V \rightarrow W$ is a linear transformation of finite dimensional F -vector spaces. Show:
 - (a) If T is injective, then $\dim V \leq \dim W$.
 - (b) If T is surjective, then $\dim V \geq \dim W$.

Definition 1. If $A \in M_{m \times n}(F)$ and $B \in M_{n \times p}(F)$ are matrices corresponding to linear transformations $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $S : \mathbb{R}^p \times \mathbb{R}^n$, then the **matrix product** $AB \in M_{m \times p}$ is the matrix corresponding to the composition $T \circ S : \mathbb{R}^p \rightarrow \mathbb{R}^m$.

- (7) If B is given as a column matrix $[v_1 \ \cdots \ v_p]$, show that AB is the column matrix $[Av_1 \ \cdots \ Av_p]$.

Definition 2. A linear transformation $T : V \rightarrow W$ is an **isomorphism** if it is invertible. In this case, by problem 2, $T^{-1} : W \rightarrow V$ is also an isomorphism. We will emphasize the isomorphism property by writing $T : V \xrightarrow{\sim} W$.

- (8) Show that $U : F^n \rightarrow V$ is an isomorphism if and only if the vectors $U(e_1), \dots, U(e_n) \in V$ are a basis.

Definition 3. Suppose that $\mathcal{B} = \{v_1, \dots, v_n\}$ is a basis for V corresponding to an isomorphism $U : F^n \xrightarrow{\cong} V$ with $U(e_i) = v_i$. Given a linear transformation $T : V \rightarrow V$, the **matrix representation for T with respect to $\mathcal{B}$** , or simply **$\mathcal{B}$ -matrix**, is the matrix $[T]_{\mathcal{B}} \in M_{n \times n}(F)$ representing the composition

$$F^n \xrightarrow[\cong]{U} V \xrightarrow{T} V \xrightarrow[\cong]{U^{-1}} F^n$$

- (9) Let $D : \text{Poly}_{\leq 4}(\mathbb{R}) \rightarrow \text{Poly}_{\leq 4}(\mathbb{R})$ be the derivative function. Find $[D]_{\mathcal{B}}$ with respect to the following bases:

(a) $\mathcal{B} = \{1, x, x^2, x^3, x^4\}$.

(b) $\mathcal{B} = \{1, x-1, (x-1)^2, (x-1)^3, (x-1)^4\}$